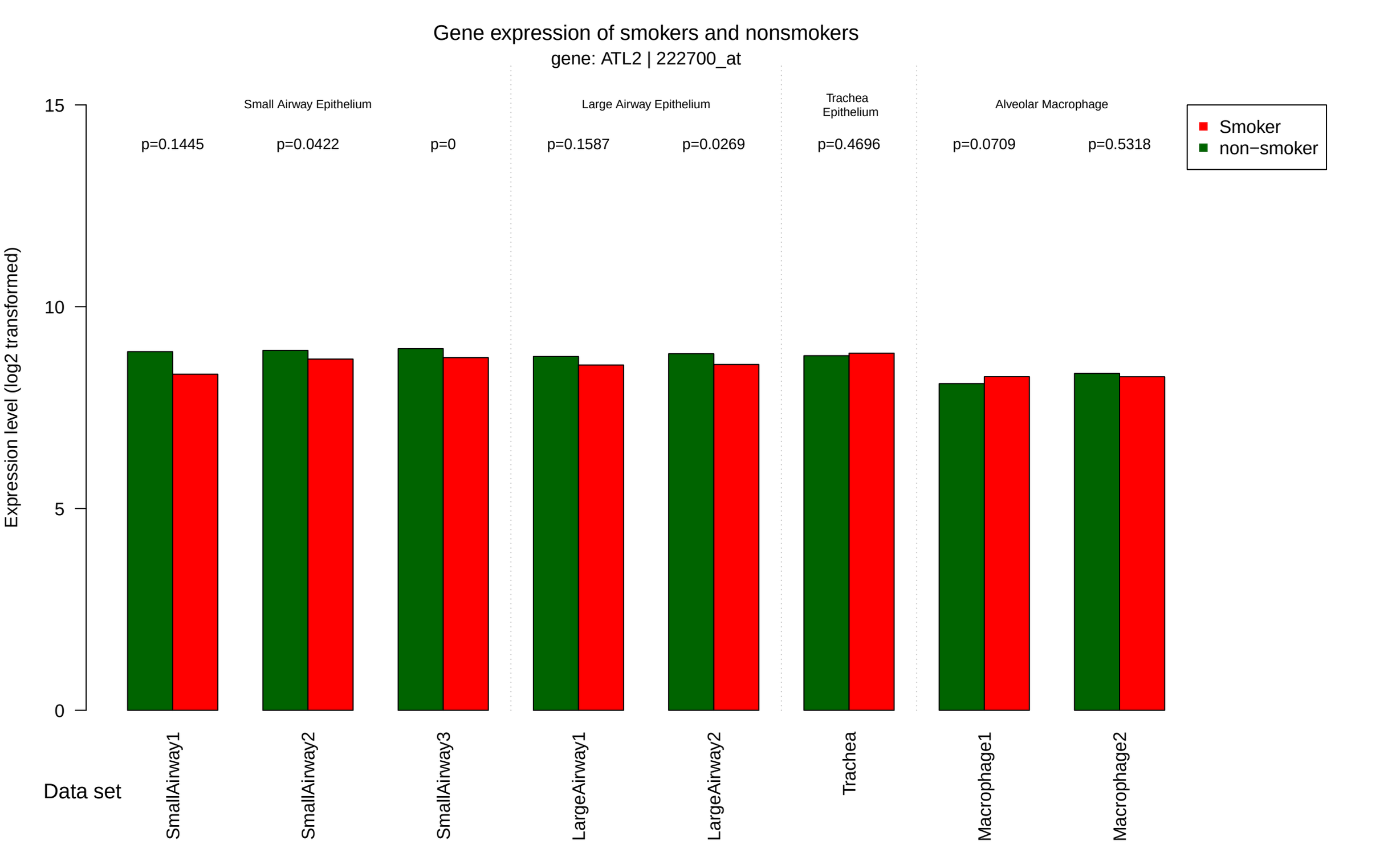
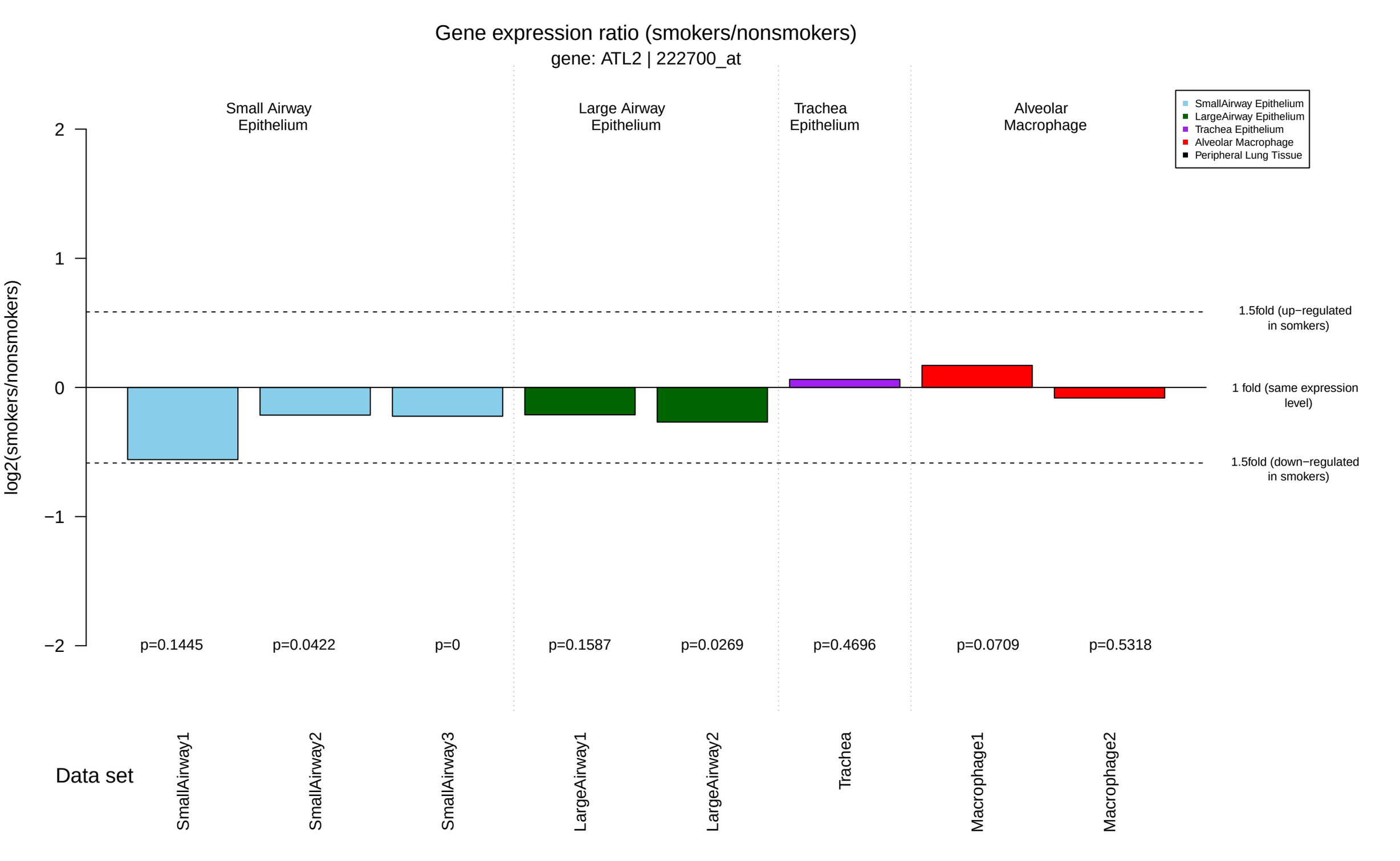
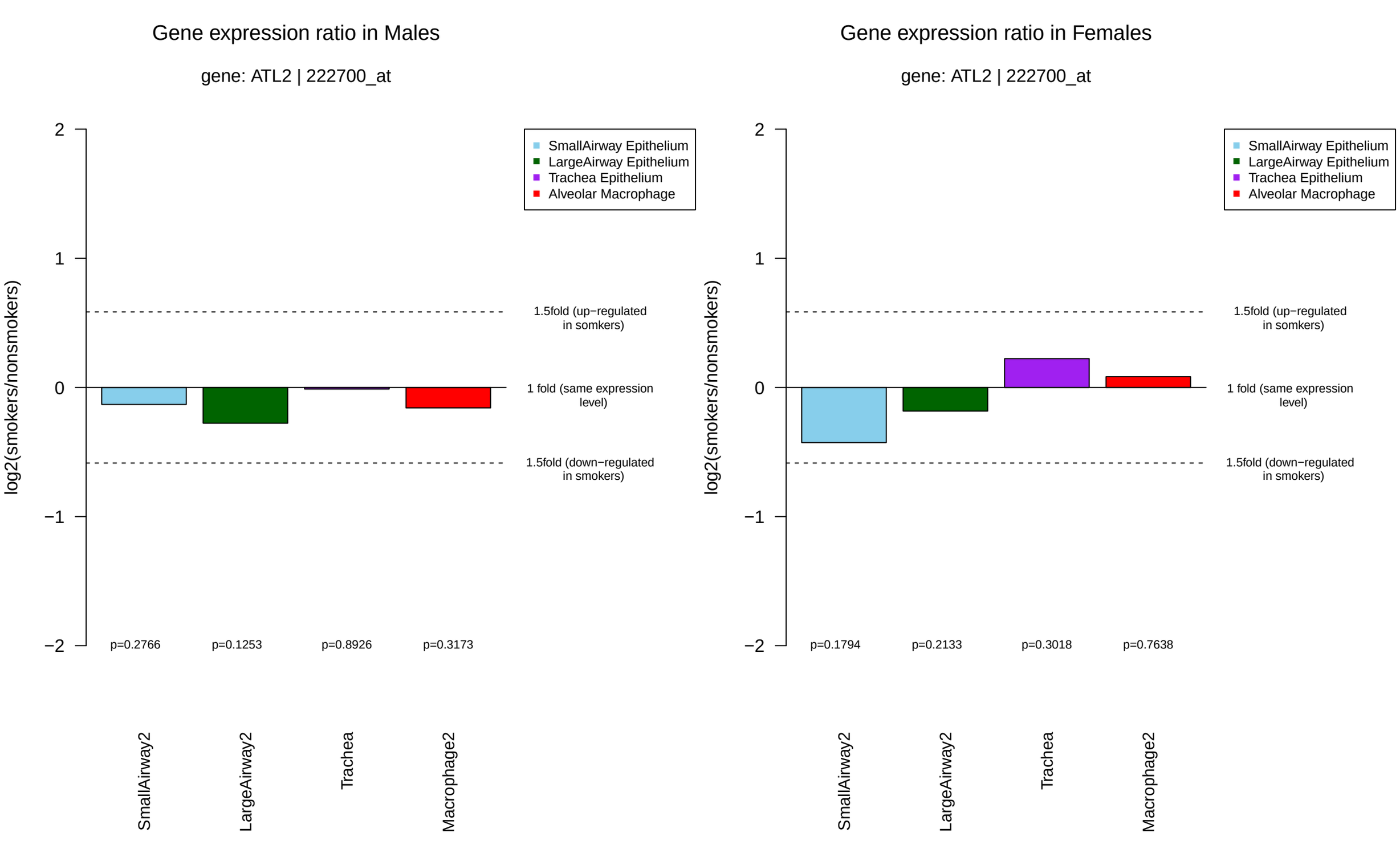
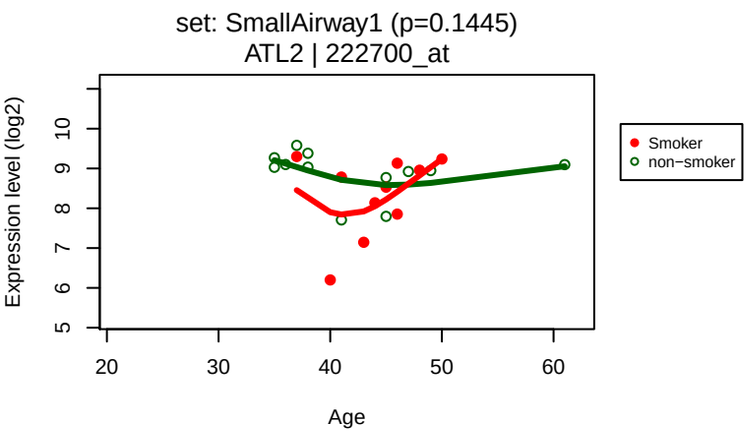
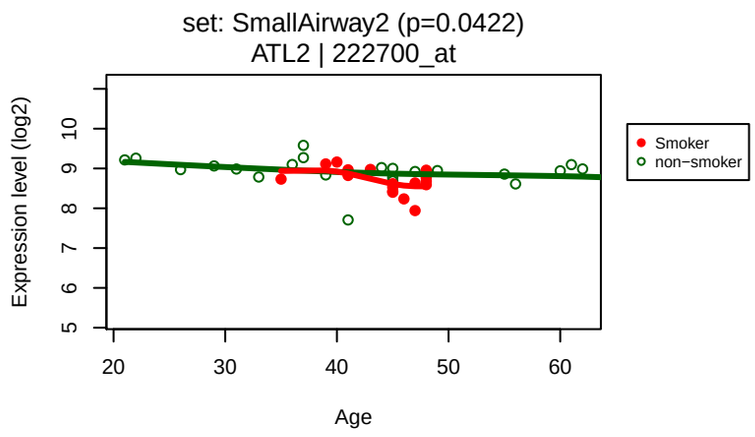
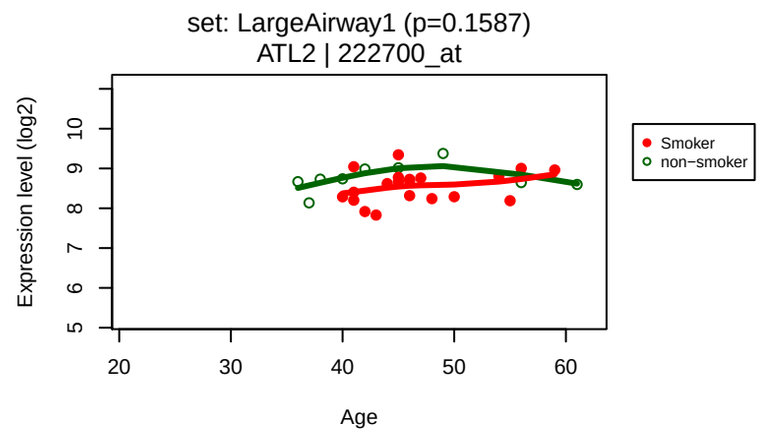

Gene expression of smokers and nonsmokers
gene: ATL2 | 222700_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1445
p=0.0422
p=0
p=0.1587
p=0.0269
p=0.4696
p=0.0709
p=0.5318
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

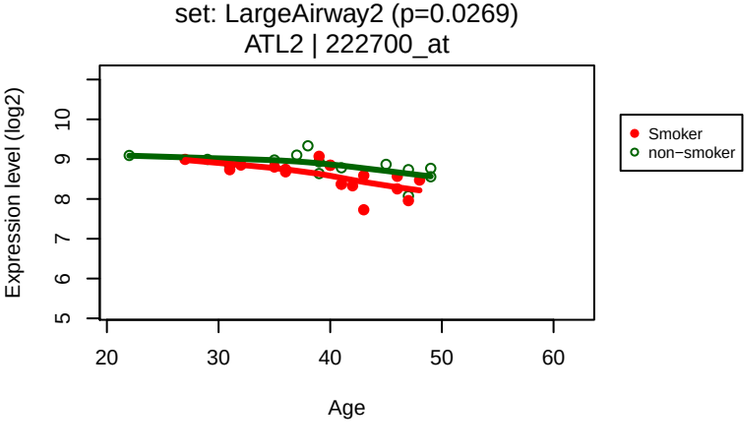

Gene expression ratio (smokers/nonsmokers)
gene: ATL2 | 222700_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1445
p=0.0422
p=0
p=0.1587
p=0.0269
p=0.4696
p=0.0709
p=0.5318
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

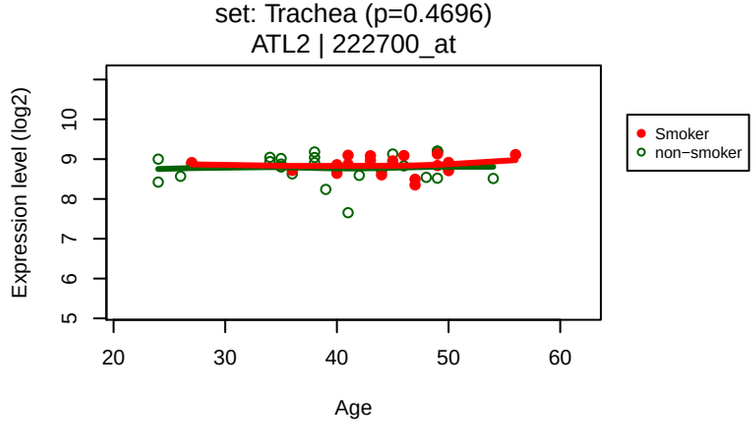

Gene expression ratio in Males
gene: ATL2 | 222700_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2766
p=0.1253
p=0.8926
p=0.3173
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATL2 | 222700_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1794
p=0.2133
p=0.3018
p=0.7638
SmallAirway2
LargeAirway2
Trachea
Macrophage2

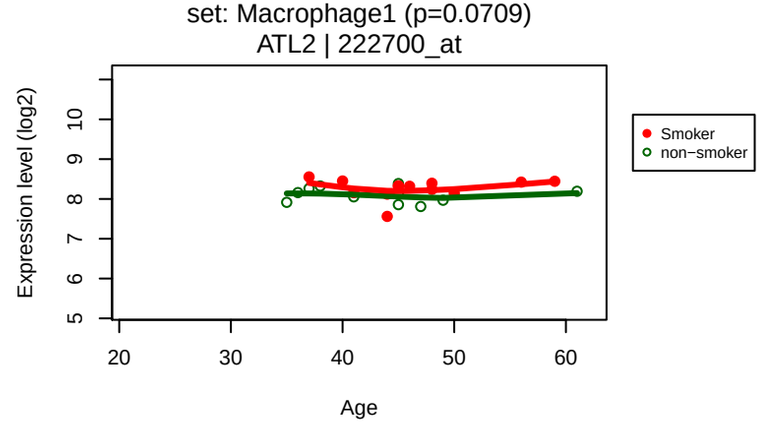

set: SmallAirway1 (p=0.1445)
ATL2 | 222700_at
Expression level (log2)
Age
Smoker
non−smoker

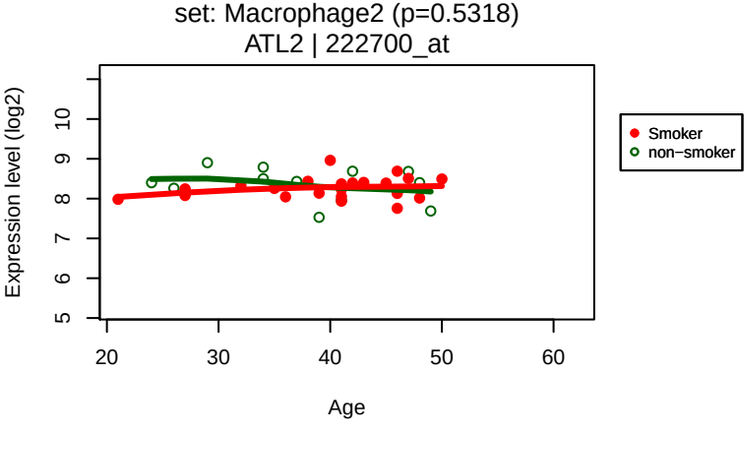

set: SmallAirway2 (p=0.0422)
ATL2 | 222700_at
Expression level (log2)
Age
Smoker
non−smoker

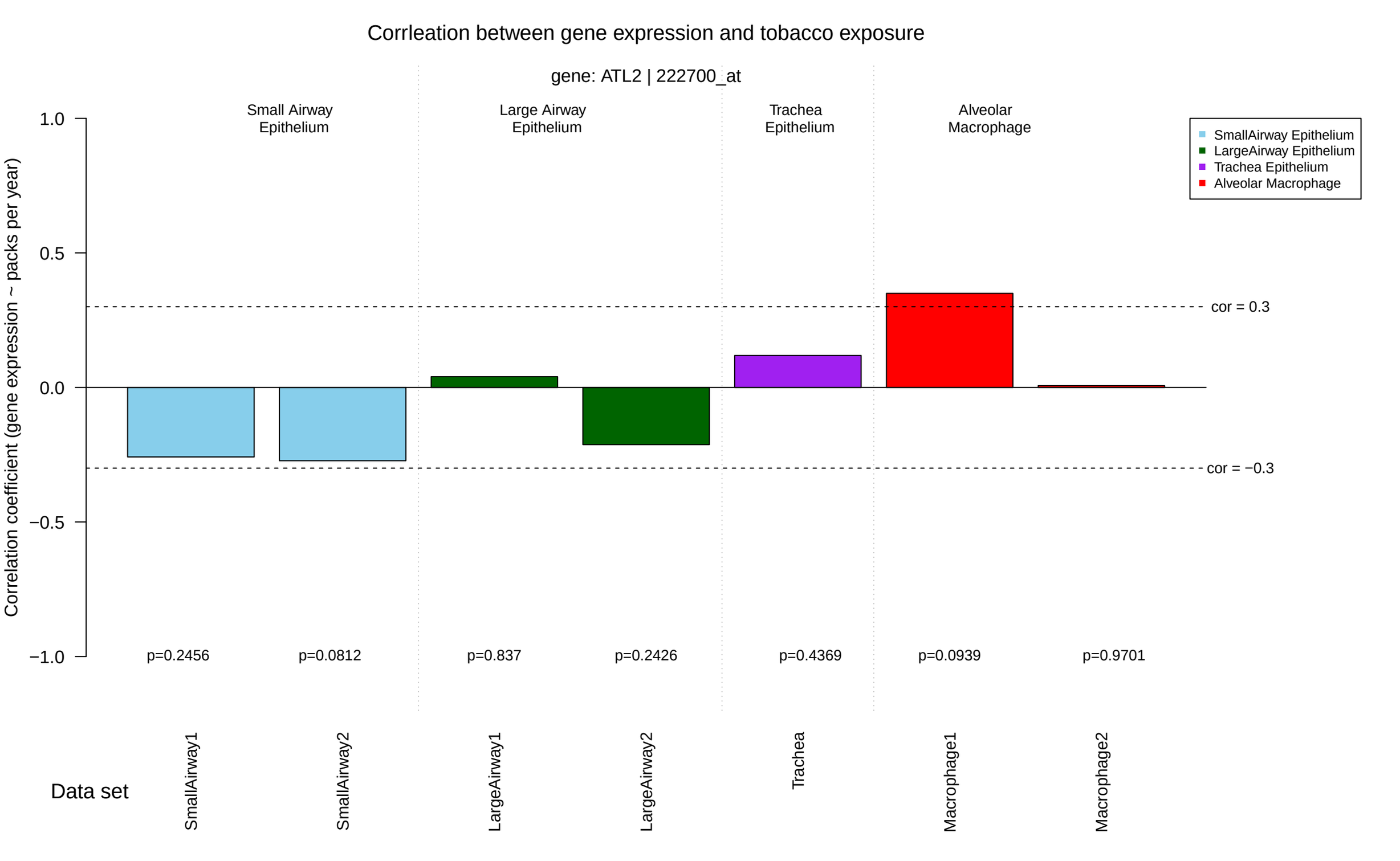

set: LargeAirway1 (p=0.1587)
ATL2 | 222700_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0269)
ATL2 | 222700_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4696)
ATL2 | 222700_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0709)
ATL2 | 222700_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5318)
ATL2 | 222700_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATL2 | 222700_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2456
p=0.0812
p=0.837
p=0.2426
p=0.4369
p=0.0939
p=0.9701
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2